



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
UNIT TEST
18-10-2020

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

Time : 3 Hours

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

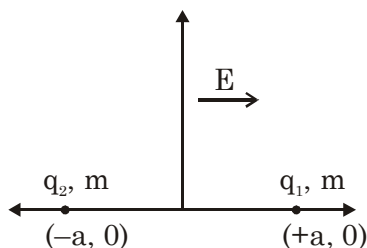
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

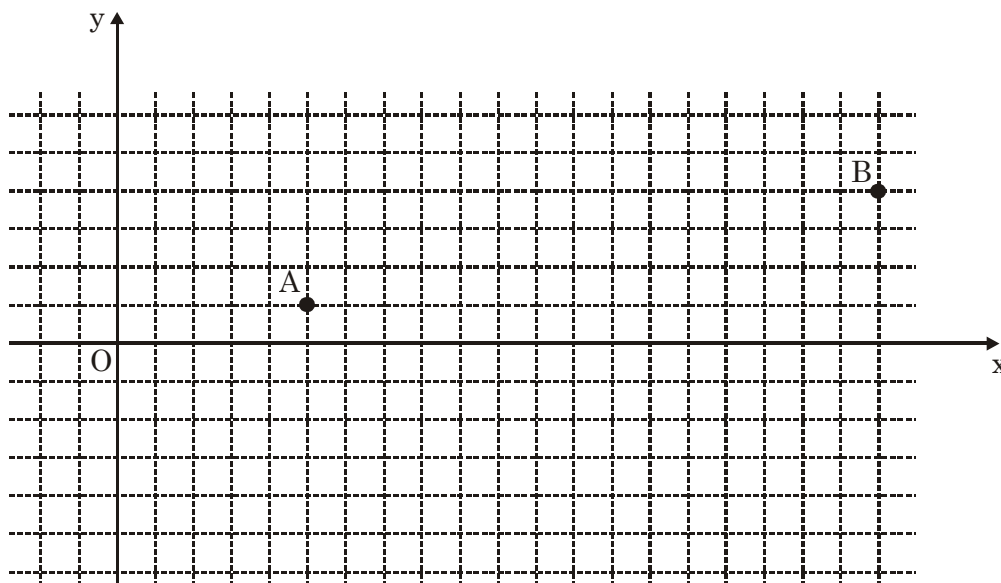
1. Two positively charged particles of charges q_1 and q_2 have mass m each. A uniform electric field E exists in positive x direction as shown in figure. Two charged particles are released from rest at $t = 0$ sec from points $(\pm a, 0)$ as shown in figure. If position of q_1 at $t = 2$ sec. is given by coordinate $(+2a, 0)$ then the x -coordinate of q_2 at $t = 2$ sec is :



- (A) $\frac{q_1 + q_2}{m} E - 2a$ (B) $\frac{q_1 + q_2}{m} E - a$ (C) $2\left(\frac{q_1 + q_2}{m}\right) E - 2a$ (D) $2\left(\frac{q_1 + q_2}{m}\right) E - a$

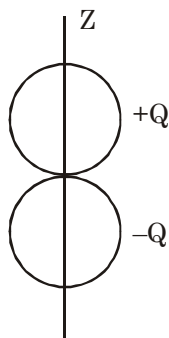
Space for Rough Work

2. Point B is the image of point A due to reflection from mirror (which is either spherical concave mirror, or spherical convex mirror or plane mirror), whose principal axis is x-axis.



One square in x-axis represents 5 cm and one square on y-axis represents 1 mm. The coordinate of centre of curvature of mirror is $(x_1, 0)$ and co ordinate of pole of mirror is $(x_2, 0)$ respectively. What will be value of $|x_1| + |x_2|$ (in cm)

- (A) 40 (B) 75 (C) 25 (D) 100
3. An electric dipole is constructed by fixing two circular charged rings, each of radius a , with an insulating contact, one of these rings has a total charge $+Q$ and the other has total charge $-Q$. If the charge is distributed uniformly along each ring, the dipole moment about the point of contact will be



- (A) $4aQ(+\hat{k})$ (B) $2aQ(+\hat{k})$ (C) $8aQ(+\hat{k})$ (D) $\pi aQ(+\hat{k})$

Space for Rough Work

4. Moment of inertia of a uniform hexagonal plate about an axis LL' is I as shown in the figure (a). The moment of inertia (about axis XX') of an equilateral uniform triangular plate of thickness double that of the hexagonal plate is (Ratio of specific gravity $\frac{\rho_t}{\rho_h} = 3$)

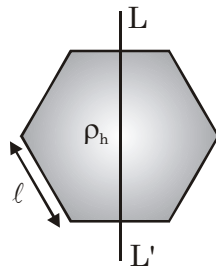


Figure (a)

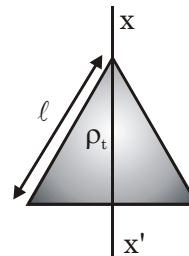
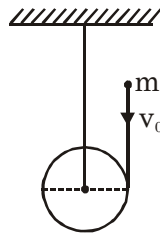


Figure (b)

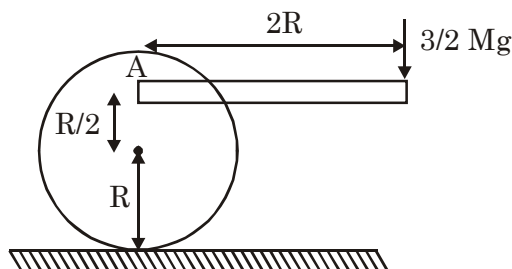
- (A) $I/5$ (B) $I/10$ (C) I (D) zero
5. A uniform disc of mass M and radius R is hanging from a rigid support and is free to rotate about a horizontal axis passing through its center in vertical plane as shown. A particle of mass m falls vertically and hits the disc (at a point at horizontal diameter) with a velocity v_0 (in m/s) and sticks to it. The disc just completes the vertical circular motion afterwards. Find v_0 (in m/s) ($M = 7$ kg, $m = 1$ kg, $R = 10$ cm).



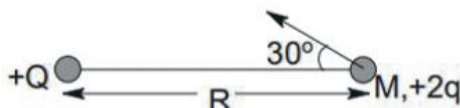
- (A) 3.5 m/s (B) 4 m/s (C) 2 m/s (D) 3 m/s

Space for Rough Work

6. A disc of mass M & radius R is placed on a rough horizontal surface with its axis horizontal. A light rod of length ' $2R$ ' is fixed to the disc at point 'A' as shown in figure and a force $\frac{3}{2} Mg$ is applied at the other end. If disc starts to roll without slipping, $\mu_{\min}$ is minimum coefficient of friction between disc & horizontal surface required for pure rolling find value $\mu_{\min}$:-



- (A) 0.5 (B) 0.4 (C) 0.6 (D) 0.8
7. In the diagram shown, the charge $+Q$ is fixed. Another charge $+2q$, is projected from a distance R from the fixed charge. The minimum separation between the two charges is when the velocity becomes $\frac{1}{\sqrt{3}}$ times of the projected v velocity, is (Assume gravity to be absent)



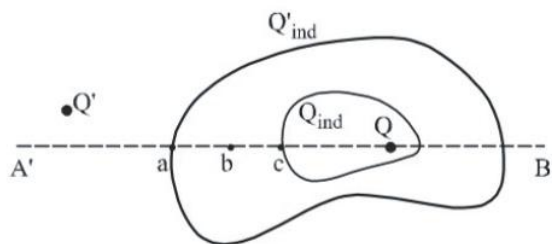
- (A) $\frac{\sqrt{3}}{2}R$ (B) $\sqrt{3}R$ (C) $\frac{1}{2}R$ (D) $4R$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

8. Figure shows a irregular shape conductor with irregular cavity inside it. A charge Q is placed inside cavity and a charge Q' is placed outside conductor. Let Q'_{ind} be charge induced at outside surface of conductor and Q_{ind} be the charge induced at inside surface of cavity. A'B' is an arbitrary line passing through charge Q and a, b, c be the points on line as shown in figure. Let $\vec{E}, \vec{E}', \vec{E}_{ind}$ and $\vec{E}'_{ind}$ represent electric field at different points due to charge Q, Q', Q_{ind} & Q'_{ind} respectively. If V_a, V_b and V_c represent potential at point a, b, and c. Choose the incorrect statement. (Assume potential zero at infinity).



(A) $V_a = V_b = V_c$

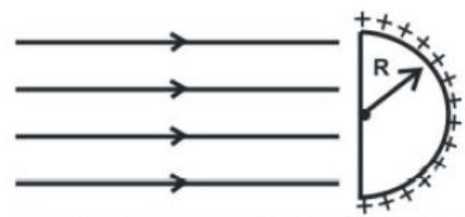
(B) $V_b = -\int_{\infty}^{\vec{r}_b} (\vec{E} + \vec{E}' + \vec{E}_{ind} + \vec{E}'_{ind}) \cdot d\vec{r}$, where $\vec{r}_b$ = position vector of point b

(C) $V_b = -\int_{\infty}^{\vec{r}_b} (\vec{E} + \vec{E}'_{ind}) \cdot d\vec{r}$

(D) At point 'b', $\vec{E}' = 0$

Space for Rough Work

9. In displacement method, the distance between object and screen is 96 cm. The ratio of length of two images formed by a convex lens placed between them is 4:
 (A) Ratio of the length of object to the length of shorter image is 2.
 (B) Distance between the two positions of the lens is 40 cm.
 (C) Focal length of the lens is 64 cm.
 (D) When the shorter image is formed, distance of the lens from the shorter image is 64 cm.
10. A semi circular disc of mass M and radius R with linear charge density λ on its curved circumference is hinged at its centre O and placed in a uniform electric field as shown in the figure. (Neglect gravity)



- (A) The net force acting on the ring is $\pi R \lambda E$ due to electric field.
 (B) If the ring is slightly rotated about O and released, find its time period of oscillation

$$\pi \sqrt{\frac{M}{\lambda E}}.$$

- (C) The work done by an external agency to rotate it through an angle α is $4\pi E R^2 \sin^2\left(\frac{\alpha}{2}\right)$.

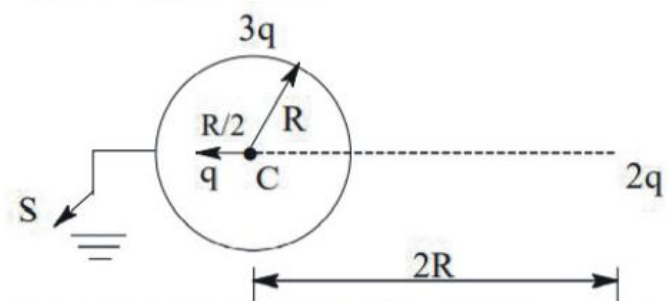
- (D) Magnitude of electric field at 'O' will be $E - \frac{2K\lambda}{R}$.

Space for Rough Work

11. A drum of mass m_1 and radius r_1 rotates freely with initial angular velocity ω_0 . A second drum with mass m_2 and radius r_2 ($r_2 > r_1$) is mounted on same axle and is at rest although it is free to rotate. A thin layer of sand with mass m is distributed on inner surface of smaller drum. At $t = 0$, small perforations in the inner drum are opened. The sand starts to fly out at a constant rate λ kg/sec and sticks to the outer drum. Ignore the transit time of the sand. Choose the **CORRECT** alternatives.
- (A) Angular speed of outer drum at time t is $\frac{\lambda t \omega_0}{m_2 + \lambda t} \left(\frac{r_1}{r_2} \right)^2$.
- (B) Difference in final angular speeds of two drums is 0.
- (C) Difference in final angular speeds of two drums is $\left(\frac{m(r_2^2 - r_1^2) + m_2 r_2^2}{(m + m_2) r_2^2} \right) \omega_0$.
- (D) Angular speed of inner drum remains constant.
12. Both ends of glass rod with an index of refraction 1.6 are ground and polished to convex hemispherical surfaces. The radius of curvature at the left end is 6 cm and at right end is 12cm. The length of the rod between the vertexes is 25 cm. The object for the surface at the left end is an arrow that lies 23 cm to the left of the vertex of this surface. The arrow is 1.50mm tall and right angles to the axis. Choose the **CORRECT** option is-
- (A) The object position for the surface at right end of rod is +3.3 cm.
- (B) Object for the surface at right end of rod is virtual
- (C) Final image is formed 1.9 cm to right of vertex at right end of rod.
- (D) Final image is real.

Space for Rough Work

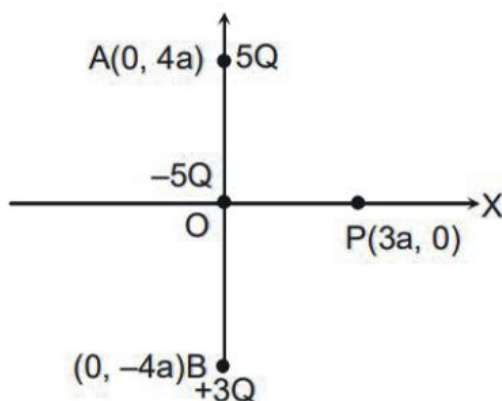
13. A spherical conducting shell of radius R is given charge $3q$ on its surface and a point charge q is placed at distance $R/2$ from its centre C . Also there is a charge $2q$ placed outside the shell at a distance of $2R$ as shown. Then



- (A) The magnitude of electric field at the centre C due to charges on the outer surface of shell is $\frac{Kq}{2R^2}$ before closing the switch S .
- (B) The electric potential at the centre C due to charges on the outer surface of shell is $\left(\frac{-Kq}{R}\right)$ before closing the switch S .
- (C) The electric potential at the centre C due to charges on the outer surface of shell is $\left(\frac{-Kq}{R}\right)$ after closing the switch S .
- (D) Charge flown through the switch in to earth after closing the switch S is $5q$

Space for Rough Work

14. Three point charges are kept at points A(0, 4a), O(0, 0) and B(0, -4a) as shown in the figure.



- (A) Work done by the external agent who assembled these charges in moving them from infinity to the point is negative.
- (B) The electric field at point P is $\vec{E}_p = E_x \hat{i} + E_y \hat{j}$, then E_x is negative and E_y is positive.
- (C) The electric field at point P is $\vec{E}_p = E_x \hat{i} + E_y \hat{j}$, then both E_x and E_y are negative.
- (D) A positive charge $\frac{975Q}{8}$ has to be put at point P to make the potential energy of system of all four point charges to be zero.

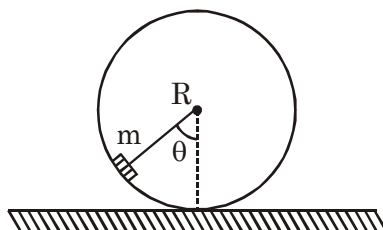
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

A particle of mass m is fixed on a massless ring which rests on a sufficiently rough surface so as to prevent slipping. The system is released from the shown configuration. Then :



15. Immediately after releasing from the shown configuration what is the angular acceleration of the ring?

(A) $\frac{g \sin \theta}{4R \sin^2 \left(\frac{\theta}{2} \right)}$ (B) $\frac{g \sin \theta}{2R}$ (C) $\frac{g \sin \theta}{R}$ (D) $\frac{g \sin \theta}{R(1 - \cos \theta)}$

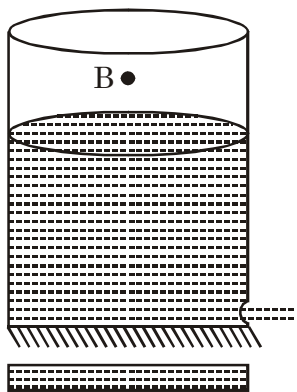
16. If α be the anticlockwise angular acceleration immediately after release from the shown position, then what is the normal reaction immediately after release:

(A) $N = mg - mR\alpha \cos \theta$ (B) $N = mg$
 (C) $N = m[g - R\alpha \sin \theta]$ (D) None of these

Space for Rough Work

Paragraph for Questions 17 and 18

In a cylindrical container of area 100 cm^2 , water is filled upto a height 80 m . there is a small orifice at the bottom of the container. The area of orifice is 1 cm^2 . The base of the container is silvered and behaves like plane mirror. An insect is crawling down on the wall of the container with velocity 2 cm/s and at $t = 0$ it is at 81 m from the base. At $t = 0$ water starts flowing out from the orifice, (refractive index of water is $4/3$)



17. The velocity of image of insect as observed by the insect at $t = 200 \text{ sec}$.
 (A) 2 cm/s (B) 6 cm/s (C) 3 cm/s (D) 8 cm/s
18. After a certain time velocity of image of insect becomes constant. The time at which this happens will be
 (A) 200 sec (B) 10 sec (C) 400 sec (D) 300 sec

Space for Rough Work

PART-2 : CHEMISTRY

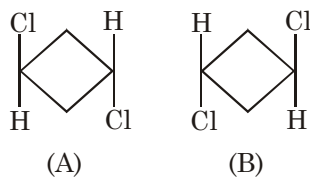
SECTION-I(i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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Negative Marks : -1 In all other cases

1. The class of non-metal chlorides often depict different properties in solid, liquid & gaseous states. Which of the following is **TRUE** for such chlorides.
(A) Atmost one chloride out of all chlorides of group 15 & group 17 exist in ionic form in solid/liquid phase.
(B) $\widehat{\text{ClECI}}$ bond angles in tetrachlorides of group-14 elements decreases down the group
(C) The bond angle decreases from top to bottom in group 15 trichlorides.
(D) The bond angle decreases from top to bottom in group 14 tetrachlorides.
2. In which of the following process, the underlined specie will not be attracted to charged comb when brought near it.
(A) $\text{HCN} \rightleftharpoons \frac{1}{2}(\underline{\text{CN}})_2 + \text{H}^+ + \text{e}^-$
(B) $\text{PCl}_3 + 3\text{LiH} \rightarrow \underline{\text{PH}_3} + 3\text{LiCl}$
(C) $3\text{HNO}_2 \longrightarrow \text{HNO}_3 + \underline{2\text{NO}} + \text{H}_2\text{O}$
(D) $\text{NH}_4\text{NO}_3 \longrightarrow \underline{\text{N}_2\text{O}} + 2\text{H}_2\text{O}$

Space for Rough Work

3.

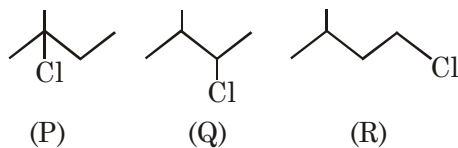


True statement for above molecules :

- (A) (A) & (B) molecules are optically active and enantiomers
 - (B) (A) & (B) molecules are optically inactive and enantiomers
 - (C) (A) & (B) molecules are optically inactive and diastereomers
 - (D) (A) & (B) are identical
4. Select correct statement(s) among the following :
- (A) Polar protic solvent favours S_N2 mechanism more than polar aprotic solvent generally
 - (B) Nucleophilic order of halides in polar protic solvent is $I^- > Br^- > Cl^- > F^-$
 - (C) Always perfect racemisation take place in S_N1 mechanism
 - (D) Inversion of configuration in S_N2 mechanism always produces R to S configuration or S to R configuration

Space for Rough Work

5. Select the incorrect order of among following :



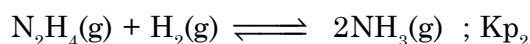
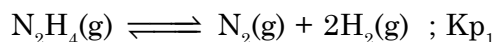
- (A) Reactivity towards E1 reaction is $P > Q > R$
 (B) Reactivity towards E2 reaction is $P > Q > R$
 (C) Reactivity towards S_N1 reaction is $P > Q > R$
 (D) Reactivity towards S_N2 reaction is $P > Q > R$
6. An equimolar mixture of $K_3[Fe(C_2O_4)_3]$ and $K_2[Fe(C_2O_4)_2(H_2O)_2]$ was oxidised completely by 66ml 0.2M $KMnO_4$ in acidic medium. Millimoles of any one compound in original mixture is -
 (A) 3 (B) 6 (C) 1 (D) 1/4
7. In what mole ratio, a mixture of N_2 and NH_3 gases should be taken such that the mole - fraction of N_2 at any instant remains constant, even after dissociation of NH_3 into N_2 and H_2 ? Mole fraction is the ratio of mole of that gas and the total mole of gases.
 (A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 3 : 2

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
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8. Hydrazine (N_2H_4) undergoes following reactions when heated at 1000 K as



In a rigid vessel $\text{N}_2\text{H}_4(\text{g})$ was introduced at 1 atm initial pressure at 1000K. Total pressure at equilibrium was found to be 1.4 atm and partial pressure of H_2 was 0.3 atm.

Select the correct option(s)

(A) Partial pressure of $\text{N}_2\text{H}_4(\text{g})$ at equilibrium is 0.7 atm

(B) P_{NH_3} at equilibrium is 0.2 atm

(C) $K_{p_1} = \frac{6}{7} \times 10^{-1} (\text{atm})^2$

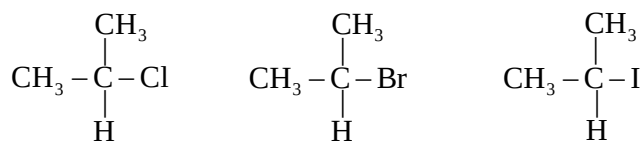
(D) $K_{p_2} = \frac{4}{21}$

Space for Rough Work

9. Correct option(s) out of the following is/are -

- (A) In FCC unit cell, fraction of edge not covered by atoms is 0.293.
 (B) In BCC unit cell, the edge length is equal to $\frac{2X}{\sqrt{3}}$, where X = nearest neighbour distance.
 (C) In FCC unit cell, distance between nearest octahedral void and tetrahedral void is $\frac{a}{\sqrt{2}}$;
 (a = edge length of unit cell)
 (D) HCP and CCP lattice of same element have same fraction of unoccupied space.

10. From left to right, correct statements are :



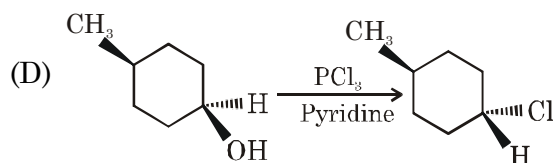
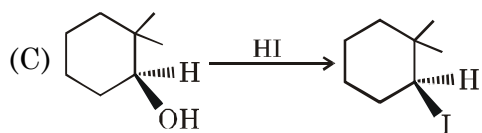
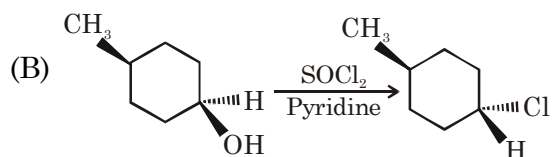
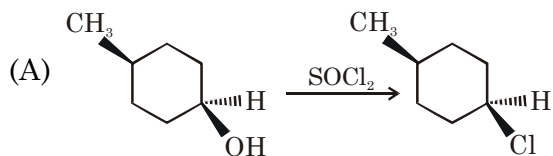
- (A) Rate of S_N1 mechanism increases in polar protic solvent
 (B) Rate of S_N2 mechanism increases in DMSO
 (C) Rate of E_2 mechanism increases
 (D) Rate of E_1 mechanism increases

Space for Rough Work

11. Which of the following statement is/are incorrect :

- (A) Optically inactive compounds can not show optical isomerism
- (B) Non resolvable compounds are optically active
- (C) Diastereoisomers are always optical isomers of each other
- (D) Mixture of diastereoisomers is always optically active

12. Which of the following reaction is showing correct major product :



Space for Rough Work

13. Which of the following is/are **INCORRECT** for hydrides of p-block?
- (A) All hydrides of group 15 do not show hybridized structure and involves pure orbital in bonding with H atom.
 - (B) The % p character in lone pair decreases from NH_3 to SbH_3
 - (C) All hydrides of second period are polar and non planar
 - (D) Hydrides of p-block may or may not show H-bonding
14. Consider an hypothetical system that has azimuthal quantum number values from 0 to n and values of spin quantum number as $\left(\frac{-1}{2}, 0, \frac{+1}{2}\right)$. (Assume all other rules to be same).

Which is/are correct.

- (A) 1st period of such periodic table will contain 8 elements
- (B) Atomic number 15 must belong to s-block
- (C) 2nd shell can contain 28 electrons
- (D) H^- will have degeneracy of 3 in 1st excited state

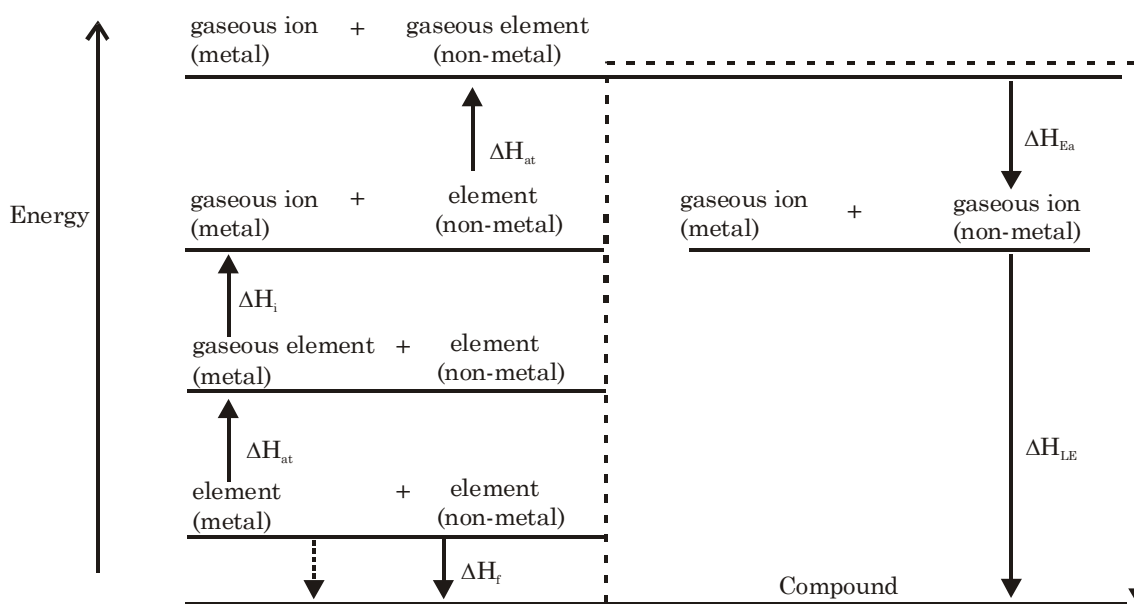
Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

The Born-Haber cycle is an approach to analyze reaction energies. A Born-Haber cycle applies Hess's law to calculate lattice enthalpy by comparing standard enthalpy change of formation of ionic compound to the enthalpy required to make gaseous ions from elements. Consider the following cycle.



15. Which of the following is **CORRECT**?
- (A) If compound in the above cycle is $MgCl_2$, then only two ΔH values will be negative
- (B) If compound in above cycle is Mg_3N_2 , then only two ΔH values will be negative
- (C) ΔH_{Ea_1} will have more negative value in case of compound NaF compared to the case of compound NaCl
- (D) All of these
16. For Born Haber cycle of AlN based on above cycle, which of the following ΔH values will be negative.
- (A) ΔH_{IE} (B) $\Delta H_{atomization}$ (C) ΔH_{EA} (D) ΔH_{LE}

Space for Rough Work

Paragraph for Questions 17 and 18

For a non-ideal gas, the compressibility factor (Z) is defined as

$$Z = \frac{pV_m}{RT}; V_m = \text{molar volume}$$

Compressibility of an unknown gas at 600K and 1.0 atm was found to be 1.2. Also this gas was found to effuse 1.58 times slower than the pure methane gas under identical conditions.

17. Density of the gas in the above mentioned experimental condition is -

- (A) 0.98gL^{-1} (B) 0.68gL^{-1} (C) 1.02 gL^{-1} (D) 1.47gL^{-1}

18. Molar volume of the gas in the given experimental condition is -

- (A) 40.8 L (B) 39.2 L (C) 58.8 L (D) 27.2 L

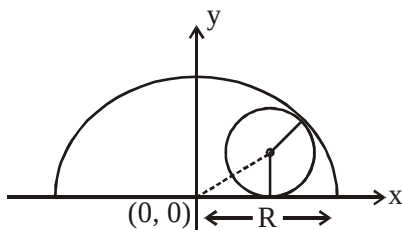
Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 21)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. Let $T_n = \left(\frac{n!}{1.3.5.7.....(2n+1)} \right)^2$, $n \in \mathbb{N}$, then sum of infinite terms of the series $T_1 + T_2 + T_3 + \dots$
- (A) does not exist (B) is less than $\frac{4}{27}$
- (C) is greater than $\frac{4}{27}$ (D) none of these
2. Let a variable circle is inscribed in a given semicircle of constant radius R as shown in the diagram. Then the locus of centre of variable circle is



- (A) Circle (B) Parabola (C) Ellipse (D) Hyperbola

Space for Rough Work

3. If the normal at any point P on the hyperbola $5x^2 - y^2 = 5$ meets the x-axis at Q and y-axis at

R, then $\left| \frac{PQ}{PR} \right|$ is-

- (A) 1 (B) 3 (C) 4 (D) 5

4. If $\left(\frac{x^2 + 2x + 2}{x + 1} \right)^{2018} = a_{2018}x^{2018} + a_{2017}x^{2017} + \dots + a_1x + a_0 + \frac{b_1}{x + 1} + \frac{b_2}{(x + 1)^2} + \dots + \frac{b_{2018}}{(x + 1)^{2018}} \forall x \in \mathbb{R} - \{-1\}$,

then $(a_i, b_i \in \text{Constant})$

(A) $\sum_{i=0}^{2018} a_i + \sum_{i=1}^{2018} \frac{b_i}{2^i} = \left(\frac{5}{2} \right)^{2018}$

(B) $\sum_{i=1}^{2018} b_i = 2^{2018} - \binom{2018}{1009}$

(C) $\sum_{i=1}^{2018} b_i = 2^{2018} + \binom{2018}{1009}$

(D) $a_0 = \binom{2018}{1009}$

Space for Rough Work

5. $\sum_{r=0}^{n-p} (-1)^{p+r} \cdot {}^nC_r \cdot {}^{n-r}C_p \cdot 5^{n-p-r} \cdot 6^r$ is equal to

(A) $(-1)^p \cdot 5^{n-p}$

(B) ${}^nC_p (-1)^n$

(C) $(-1)^n \cdot 5^{n-p}$

(D) ${}^nC_p (-1)^p$

6. A person wants to climb a n -step staircase using one step or two steps. Let C_n denotes the number of ways of climbing the n -step staircase. Then $C_{18} + C_{19}$ equals

(A) C_{20}

(B) C_{21}

(C) greater than C_{21}

(D) less than C_{20}

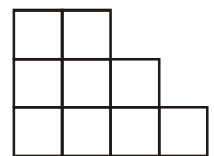
7. The number of ways in which 5 letters of word ALLEN can be placed in the squares of the figure so that no row remains empty is -

(A) 98

(B) 5880

(C) 7560

(D) none of these



Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 28)

- This section contains **SEVEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

8. If ΔABC with usual notations $r = 2$, $r_1 = 4$, $s = 12$ and $a < b < c$. Identify the correct statements
- (A) $\Delta = 24$ sq. units (B) $R = 6$
- (C) $|r_2 - r_3| = 6$ (D) least angle of triangle is $\sin^{-1}\left(\frac{3}{4}\right)$
9. Possible value(s) of a for which the lines $2x + y - 1 = 0$, $ax + 3y - 3 = 0$ and $3x + 2y - 2 = 0$ are concurrent is/are-
- (A) 1 (B) -1 (C) 2 (D) -2

Space for Rough Work

10. Loci of a point $P(x, y)$ satisfying the relation $x^4 + y^4 + 2x^2y^2 - 2x^2 - 6y^2 + 1 = 0$ are the two circles, these circles
- (A) have centre on y-axis (B) cut each other orthogonally
(C) are congruent (D) have exactly two common tangents.
11. Let two parabolas have a common axis where focus of each being exterior to the other and let ℓ_1, ℓ_2 be their latus rectums then the locus of the mid points of the intercepts between the parabolas made on the lines parallel to the common axis is a
- (A) Straight line if $\ell_1 = \ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$ (B) Parabola if $\ell_1 \neq \ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$
(C) Parabola $\forall \ell_1, \ell_2 \in \mathbb{R}^+$ (D) Hyperbola if $\ell_1 = 2\ell_2 \forall \ell_1, \ell_2 \in \mathbb{R}^+$
12. From a point on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, two tangents are drawn to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the corresponding chord of contact meets $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ at points Q and R then
- (A) The locus of mid-point of QR is $x^2 + y^2 = a^2$
(B) The locus of mid-point of QR is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
(C) The locus of image of mid-point of QR in x-axis is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
(D) The locus of foot of perpendicular from point $(\sqrt{a^2 + b^2}, 0)$ to QR is $x^2 + y^2 = a^2$

Space for Rough Work

13. Let $(2 + x) (1 + (1 + x)^2) (1 + (1 + x)^4) (1 + (1 + x)^8) (1 + (1 + x)^{16}) = a_0 + a_1x + \dots + a_nx^n$, then

(A) $a_r = a_{n-r}$, for each $r = 0, 1, 2, \dots, n$

(B) $a_{14} = a_{16}$

(C) $\sum_{r=0}^n a_r = 2^{32} - 1$

(D) $a_0 + a_n = 2$

14. Consider the set A of all determinants of order 3 with entries 0 or 1 only. Let B be the subset of A consisting of all determinants with value 1. Let C be the subset of A consisting of all determinants with value -1, then

(A) $C = \phi$

(B) $A = B \cup C$

(C) $B \cap C \neq \phi$

(D) Number of elements in B and that in C are equal

Space for Rough Work

SECTION-I(iii) : (Maximum Marks : 12)

- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

Paragraph for Questions 15 and 16

A Rod of length 10cm slides between two lines, x-axis & $y = -\sqrt{3}x$. If locus of mid point of rod is curve 'S'.

15. Which of following statement is INCORRECT ?

- (A) S is an ellipse (B) Centre of S is (0,0)
(C) x-intercept of S is 10 (D) y-intercept of S is $10\sqrt{3}$

16. Maximum distance between origin and curve S is-

- (A) $5\sqrt{3}$ (B) $5\sqrt{2}$ (C) 5 (D) $\sqrt{5}$

Space for Rough Work

Paragraph for Questions 17 and 18

A cricket team of eleven players is to be selected out of 6 batsman, 5 bowlers, 2 all rounders and 2 wicket keepers. Consider an allrounder as a player who is a batsman as well as a bowler.

17. The number of selections containing exactly one wicket keeper and atleast one all rounder is-

(A) ${}^2C_1(2 \cdot {}^{11}C_9 + {}^{11}C_8)$

(B) ${}^2C_1 \cdot {}^2C_9 \cdot {}^{11}C_9$

(C) ${}^2C_1 \cdot ({}^{12}C_3 + {}^{11}C_3)$

(D) ${}^4C_2 \cdot {}^{11}C_9$

18. Number of selections when a particular wicket keeper do not want to play if a particular batsman and a particular bowler are both playing, is (where it is not necessary that a wicket keeper must be a part of playing eleven)

(A) ${}^{13}C_{11} + {}^{13}C_{10} + {}^{12}C_9$

(B) ${}^{15}C_{11} - 2 \cdot {}^{13}C_3$

(C) ${}^{13}C_{11} + 2 \cdot {}^{13}C_{10} + {}^{12}C_9$

(D) ${}^{13}C_{11} + {}^{13}C_{10} + {}^{12}C_8$

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has one section as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	7	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	21
I(ii)	One or more correct option(s)	7	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	28
I(iii)	Paragraph Based (Single correct option)	4	+3 If only the bubble corresponding to the correct option is darkened	—	0 In all other cases	—	12

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator